

Review

When Will Controlled Environment Agriculture in Its Vertical Form Fulfill Its Potential?

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Abstract: Food systems around the world are challenged to meet increased demand while also mitigating ecosystem pressures from their current structure. Controlled environment agriculture (CEA) offers a potential solution to augment the food supply by adopting innovative production systems designed to overcome environmental resource limitations and efficiently serve densely populated urban areas. By utilizing Elkington's profit, plant, and people framework (3Ps), this article assesses the sustainability of a major subcategory of CEA farms: indoor agriculture vertical farms (IA/VFs). The qualitative analysis attempts to answer the question of whether IA/VFs have fulfilled their potential. Results suggest that IA/VFs have not yet optimized their positive impact on future food system sustainability. For each of the three Ps, IA/VF's observed progress and required breakthroughs are summarized. Notably, the financial viability of an IA/VF is more likely to be achieved through whole systems solutions: growing the right crops in the right environment, efficient use of resources, and effective consumer targeting. Significant progress is being made in the direction of innovating IA/VF's role in future food systems. Through public-private partnerships and further analyses, further progress can be made toward realizing IA/VF's potential to address the growing demands of an expanding world population and shrinking resource base.



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1. Introduction

Controlled environment agriculture (CEA), particularly in its vertical farming form, has been a subject of research and industry experimentation for at least 25 years. Despommier is often cited as formally defining the concept [1]. Despommier's lab started developing a vertical farm as early as 1999 [2]. CEA has been studied, implemented in various farming configurations, and debated as to its efficacy ever since. The motivation for and its potential can be distilled into the following proposition.

As the human population grows, which is expected to reach 10 billion people over the next three decades, along with intensive urbanization [3], it has become clear that global food production will need to be more environmentally sustainable than current practices demonstrate. Agriculture around the world is challenged to meet increased demand while also mitigating the myriad ecosystem pressures that cascade from food systems as they are currently structured. By 2050, population growth and eating trends are expected to

increase environmental pressures caused by food systems by an estimated 50–92% [4]. The dilemma is to best address these unsustainable trends while still increasing the amount of food grown globally. CEA, particularly in its vertical farming form, has the potential to address this dilemma by adopting innovative production systems specifically designed to overcome land and water limitations, utilizing complex environmental controls, and enabling high-density food production in limited spaces in fully enclosed vertical structures to efficiently serve densely populated urban areas.

In plant sciences, a plethora of research exists on the various technological aspects of the CEA process, including inputs, environmental controls, and farming configurations. For example, ideally designed vertical farming systems controlling lighting spectrum, intensity, and duration, as well as other environmental factors such as CO₂ level, air temperature, and humidity, can enhance the appearance, taste, or nutrient levels of leafy greens [5,6]. Further research has explored CEA's potential to address the food demand in urban areas while promoting sustainability [1,7–9]. Industry development has been supported by multidisciplinary research focused on optimizing environmental variables to maximize yield or resource use efficiency [5,10–14] and market segmentation and acceptance studies [15–19]. From the industry's perspective, particularly in the U.S., securing funding for capital-intensive vertical farming structures, specialized labor training, and managing high electricity consumption remain significant challenges [20]. However, these obstacles have not dampened the industry's overall optimism [21]. Over the past decade, numerous global conferences, such as Indoor Ag-Con and CEAg World, have been established to convene multidisciplinary stakeholders from academia, industry, and policymaking. These platforms have facilitated the development of strategic partnerships aimed at advancing global solutions, such as the CEA Alliance, CEA Food Safety Coalition, and Urban AgriTech [21].

This body of research tends to focus on piecemeal advances in various components of vertical farming but largely lacks whole system analyses, particularly focused on the socioeconomic aspects of CEA [22]. In practice, the industry of suppliers and farms has expanded dramatically [23]. However, the realization of successful farm ventures and efficiently controlled and profitable internal growing environments has remained elusive in many cases [20,21]. While CEA technological advances are highlighted in the research and economic best practices are available to supply business goals, many practitioners consistently face challenges around integrating CEA into existing food system supply chains. Problems such as unknown consumer perception, the ongoing financial burden of large capital requirements, and suboptimal efficiencies in complicated logistics hamper CEA's natural growth, which would likely stem from those technological improvements. All stakeholders—growers, buyers, consumers, and other players in the food value chain—would benefit from a systems analysis of where CEA can contribute to resilient supply chains that can feed the demand of a growing population.

Enough time, research, and experimentation have now passed to suggest that achieving CEA potential is perhaps overdue. This paper's research question becomes the following: Why has CEA had only limited impact on sustainable food systems given its claimed potential, multiple years of attention, active implementation, and hype across academia and industry? This paper hypothesizes that even though significant progress has been made, CEA has not reached its potential because of significant remaining challenges to optimize technological systems and assure sustainability (profitability, environmental impact, and social impact). This paper will assess what specific challenges exist and what breakthroughs are needed to move CEA forward. Elkington's triple-bottom-line framework is used to guide the assessment.

2. Industry Context

Since the first development of the technology, the acronym CEA has been adopted and transmuted by many authors in the literature [24], as well as many working in the CEA industry, all attempting to add clarity to what is a very complex and differentiated technology. Many CEA techniques have even been adapted by the existing greenhouse industry, thus further muddying what is meant by CEA as a distinct system. This paper focuses on a grouping of CEA farms that can be classed together according to the following criteria:

1. An enclosed structure completely separated (to the extent possible) from its external environment;
2. The sole use of artificial lighting;
3. Space optimization based on vertical structures;
4. Critical control of the internal environment for efficient use of resources.

One category of farms meeting these criteria has been named PFAL (Plant Factory with Artificial Lighting), with most PFALs located in Japan [8]. Professor Kozai is acknowledged as the leading scholar in this area. PFALs have a mostly standardized farm configuration and technology for growing leafy greens, advancing now towards strawberry production. However, they are not the only examples; many U.S. and other global vertical farms meet the criteria while using differentiated formats. For the assessment of CEA's progress in reaching its potential, this article adopts the term indoor agriculture/vertical farms (IA/VFs) used by Mitchell [24] as a more comprehensive term for farms meeting the four criteria.

Many benefits are claimed for IA/VFs:

- Contributing to solving global food, environment, resource, and social issues;
- Resource-saving and minimizing emissions on their plant supply chain from production to consumption;
- Environmentally protective through minimized pollution;
- Maximizing resource use efficiency, including conversion and energy use;
- Weather-resilient as a production system, resulting in consistent year-round yields;
- Vigilant about environmental health and welfare;
- Creating expansive employment opportunities;
- Exhibiting a living, evolving, responsive systems design with a focus on the development of standardized systems to facilitate international collaboration [25].

While these benefits are ideal, the industry itself has often wandered away from the ideal in pursuit of the possible. Much about the progress of IA/VFs has been discussed in the literature [9,26–29], but none have evaluated IA/VFs utilizing a comprehensive framework.

The lack of a unifying nomenclature and analysis for the IA/VF industry has also led to a fractured understanding of whether these farms can be successful from a business perspective. While IA/VF farms face their share of challenges, existing successful examples demonstrate business success across three strata: profit, people, and planet.

Mature start-ups like Local Bounti and Square Roots have patiently scaled up operations in collaboration with value chain partners, implementing strategies such as co-locating farms with retail locations or retailer distribution centers [30,31]. This innovative approach to logistical concerns serves two purposes. In a manner not unlike urban farms, these farms cut down on the distance between cultivation/harvest and the end consumer, therefore minimizing logistics costs (a profit-friendly tactic) and emissions (a planet-friendly tactic). This trend is also being adopted across Europe, notably in France [32].

Other international examples have shown that there are different ways to be successful as an IA/VF farm, presuming the market demand is there. Japanese plant factories, such as

those analyzed by Kozai [8], show us a few ways to tap into the market demand for clean greens. These concerns primarily grow leafy greens for cooking or salad consumption and vary in the markets they serve, from local residents to scaled grocery store distribution [25]. Another example is Pink Farms, an IA/VF in Brazil, which is growing robustly [33].

Given the going concerns' ability to be successful in ways that span financial, environmental, and sociological criteria—a triple bottom line—the evaluation of how IA/VF, as an industry, performs when viewed through the triple bottom line lens is a helpful tool.

3. Analytical Review Framework: Triple Bottom Line

The profit, planet, and people framework [34], sometimes called the 3Ps or 'triple bottom line', offers a qualitative yet comprehensive approach to this initial attempt to assess the progress of CEA in its IA/VF form to meet its sustainability potential. Originally designed as a business strategy that, when implemented, aimed to simultaneously benefit the company, its consumers, and the environment, the triple bottom-line analytical tool has gained popularity as a framework for evaluating entities, both small and large, from projects to industries [35–37]. The components of Elkington's three Ps framework can be summarized as follows:

- **Profit:** the first P considers whether the entity generates sufficient economic returns to justify its continued operation and capital investment, aligning with principles of profitability and value creation in economic decision-making. In other words, the entity targets goals and deliverables regarding consistent or predictable profits to the bottom line;
- **Planet:** the second P analyzes the environmental impact of an entity and evaluates its relative positioning within the competitive landscape, how compliant that company is with environmental regulations, and identifies strategic opportunities for reducing its environmental footprint to enhance sustainability and long-term resilience;
- **People:** the last P refers to the impact on all stakeholders in a business, from employees to customers, investors, community members, and more. The evaluation of the people factor performance can encompass the exchange of labor for wages, the social-economic impact on individuals that surround the entity, or even upstream value-chain implications, such as suppliers' labor standards and externalities.

Sustainability arises from strategies that result in positive enhancement of all three Ps, rather than merely trading off one P at the expense of the others. The next three sections of the article lay out an assessment of IA/VFs based on their impact on each of these sustainability components. The assessment is based on findings drawn from a literature review of 83 articles, including research conducted for a USDA grant entitled Improving the profitability and sustainability of indoor leafy-greens production. Table 1 organizes the 83 articles reviewed into the 3P categories plus those used to construct an overview of IA/VF as a concept, farm structure, and industry. A brief description of identified critical factors is also provided for each category. Additional tables are used at the end of the next three sections to provide an overall assessment of current progress for IA/VFs as related to the relevant P and to recommend what breakthroughs are needed to move IA/VFs toward greater sustainability.

Table 1. Overview of reviewed research articles and identified critical factors in the profit, people, and planet analysis.

Theme	Critical Factors Identified	References
IA/VF Overview (17 references)	• These references establish the general definition and technical configuration of IA/VF as an innovative technology with various potential 3P impacts, including production structure, technical advances and relevance, potential food system contributions, and urban supply chain impacts. • Several references contrast IA/VF with traditional agriculture. • Five references are specifically literature reviews about IA/VF in general with focuses on potential benefits. • Only one reference using 3P analysis.	[1,2,8,22,24,28,29,31,38–46]
Profit (41 references)	• These references focus on aspects of production efficiency and tradeoffs among physical inputs and outputs. • 19 references did some form of profitability analysis that includes economic variables and not just physical variables. • Only six references examine consumer preferences, critical to the revenue component of profitability analysis. • Five references examined supply chain issues. • Only one reference provided an economic analysis of a total IA/VF system. Other references only analyze certain components of processes and not whole systems.	[5–7,9–21,23–25,27,31,33,38,40,43,47–62]
Planet (34 references)	• These references address aspects of IA/VF planet impact. • 19 references focus on broad environmental impacts. • 14 focus on resource efficiency as critical to planet, especially energy and land use. • Five focus on climate change specifically.	[1,2,8,9,12,18,22,26–28,32–34,39,41,42,44,45,48,52,53,59,63–74]
People (34 references)	• These references address some aspects of IA/VF's impact on people, albeit without providing a unifying definition of IA/VF's social impact. • 12 references examine consumer preferences for IA/VF products, giving some insight into consumer utility benefits. • Seven references focus on food security and availability, primarily in urban settings. • Six references focus on urban policy issues to support IA/VF development. • Four references examine multiple social impacts, with only one providing a literature review. • Three references discuss job creation. • Three references discuss nutrition and health.	[3–7,10,15–19,21,22,24,29,30,32,40,46,56,60,63,65,66,68,74–82]

4. Triple Bottom Line Analysis

4.1. Profit: Financially Sustainable for IA/VF

The IA/VF industry has experienced significant economic struggles in recent years. Many larger firms have gathered substantial amounts of capital only to fail to achieve profitable returns [21]. Conversely, growers continue to operate IA/VF farms in many different environments globally, with some consistent financial successes that could be hallmarks of IA/VF's place in food systems of the future [38]. Profitability issues, each of

which are addressed below, can be traced to several critical challenges: effective capital management; highly technical control environments working to potential; limited crop selection driven by technology and its cost rather than market circumstances; IA/VF offerings often not meeting the market's demand with clear economic value; effectively scaling operations in pursuit of cost reductions.

4.1.1. Capital Management

Building an IA/VF operation entails significant capital expenditure, especially when considered alongside the cost of initiating a traditional, soil-grown farm. Capital expenditures can include the cost of land-growing technology, including racking systems, fertigation controls, environmental monitoring equipment, lighting, packaging, seeds, labor, and energy, all of which add up to make an IA/VF an expensive cultivation endeavor [63]. The land cost of building a vertical farm in an urban area can often outweigh the cost of peri-urban or rural land by a factor of 50; however, if yields are significantly better than field-grown, eventually, the IA/VF farm will compensate for the high start-up cost of land [40]. These up-front costs may be relatively fixed, yet they are controllable both from a managerial and a financial return perspective. Previous research has calculated that the starting size of a farm that maximizes the contribution margin can be quite small, allowing for profitable installation in urban areas [14].

The variable and incremental costs of running a complex controlled environment farm appear to be difficult to manage profitability. Labor, the most significant of those variable costs [47], is managed in very different ways across organizations and in different economic environments. A partial budget analysis has shown that labor expense has the greatest effect on profitability and margin contribution [14]. Given that labor comprises 41% of the variable costs in an IA/VF operation, an increase of one percent in wages resulted in an increase of one dollar in variable operating costs. This is closely followed by energy, for which a one percent increase resulted in a 0.09% increase in costs [14]. Other variable operating costs, such as packaging, make a much smaller contribution to profitability. The overall growing costs of IA/VFs are significant, interrelated, and impactful to contribution margin, which may affect crop selection and go-to-market strategies.

The variables with the biggest impacts on margin, and therefore profitability, have been identified and are being actively managed in many IA/VF operations that are striving for or have already achieved profitability. Armed with this knowledge, more IA/VF farms could be financially sustainable entities that contribute to a functional food system.

4.1.2. Achievement of the Controlled Environment

Much of the current capital in IA/VFs, including the financial investments made to high-tech farming start-ups, is sunk into very technologically enabled environmental control systems. At first blush, this investment choice of environmental control systems is relatively clear: the more controlled the environment of an IA/VF farm, the better the grower can control various input costs (such as the energy costs to power lights and complex heating, ventilation, and air conditioning (HVAC) systems, or the water utilized for plant fertigation) and therefore manage the daily financials of their farm. However, if these control systems are not utilized effectively, farms are stuck with expensive systems that fail to create cost efficiencies.

Aspects of the controlled environment challenge have been analyzed, chief among them energy use [41]. Specifically, the ability of IA/VF operations to efficiently and cost-effectively manage the energy used to power the indoor environment is largely applied to lighting the plants. Lighting energy efficiency is possible with optimal system management [14,48], and we do see growers accomplishing this today. In addition to managing

energy use through lighting manipulation, the critical insulation and ventilation of facilities (sometimes generalized as HVAC), the use of renewable energy and distributed power grids, as well as the implementation of smart systems to manage crop production and harvests, could all contribute significantly to better financial management of IA/VF facilities [64].

As evidenced by the abundance of technological solutions that have burst onto the scene in recent years [49,65], science is ready to critically control complex agricultural environments like IA/VF farms. In fact, the efficiencies gained from the rise of cannabis markets have paid dividends to indoor growers of any crop with consistently improving grow technology [50]. However, many growers are failing to maximize the potential efficiency of their controlled environments due to, in many cases, a lack of experience or understanding. This is a case of science being ahead of business, and it would behoove the industry to invest in better grower training to improve the efficiency of controlled environments.

4.1.3. Crop Selection

IA/VF growers tend to focus on short cultivation cycle crops to maximize yield and facility turns per year in the high-cost controlled environment with a moving target of variable operating costs. This is the reverse of typical crop selection in traditional agriculture, which is based on well-defined commodity categories that have known market size and price points that allow producers to make selection decisions that lead to feasibly productive crops in available fields. In addition, IA/VF farms often consider the market's demands secondarily to the 'science experiment' of growing any feasible crop in a high-tech indoor environment. As a result, a limited selection of applicable crops has been proven technically feasible on a commercial scale in IA/VF grow operations.

Leafy greens stand as the most popular crop in the industry, with their production systems widely reported in the literature [42,51,52]. Other successful crops include microgreens and herbs and high-value fruiting crops, such as strawberries, that easily fall into the category of so-called clean, green, and gourmet (CGG) produce, allowing the specific attributes of IA/VF-grown crops to garner some of the highest prices in certain segments of the market [40]. Seminal research identified a market segment in the U.S. willing to pay a premium for indoor-produced, high-quality leafy greens based on their knowledge of the indoor agriculture industry, showing there is consumer interest if there is understanding [18].

Some researchers have also begun to view IA/VF systems as ideal for growing crops that can be used in pharmaceutical manufacturing and food processing. One such example is marjoram, a culinary herb whose oil distillate is used in food manufacturing, cosmetics (as a perfumery oil), and as an insecticide/herbicide. By using multiple grow levels and LED lighting, indoor crop growth of marjoram can be continuous year-round, enabling up to three harvests during a period equivalent to a traditional field growing season—a significant and profitable increase [53]. This example of a crop type, while emergent, shows that the controllable aspects of vertical farms create a unique economic opportunity compared to traditional agriculture. The point of crop diversification as a viable financial strategy exists in traditional agriculture [54] and has yet to be applied as best practice in IA/VF. If the industry could critically approach crop selection with an eye for efficiency as well as market tolerance, these operations could have a significantly better chance of success.

While few crops have been proven profitable when grown in critically controlled environments, the research cited here is promising that with the right product, IA/VF environments can not only be sufficient but positively beneficial to growing a quality, efficient, high-yielding crop. In practice, however, many IA/VF farms do not prioritize the matching of crops grown to market demand with a profitable cost structure.

4.1.4. Consumer Acceptance of IA/VF

Increasing yield through intensive growing methods will allow for food systems to meet known accelerating demand, yet that supply of produce can offer different attributes from crops currently offered in the market. IA/VF products could be different varieties with enhanced taste, quality, or shelf life, different price points, or displayed in different packaging with different attributes at the forefront. As a result, an assessment is needed to evaluate whether consumers will purchase IA/VF products with the same or greater scrutiny than their market-dominating competition of field-grown produce.

Given the differences between conventional and IA/VF growing methods, it follows that IA/VF-produced products may have distinguishable attributes to buyers, including seasonality, coloration, and flavor [55]. This difference in attributes could create a gap in knowledge on behalf of produce consumers, leading producers and retailers to question if those same consumers understand or care about how their produce is grown. The research shows that consumers do debate the merits of controlled environment farms, including the high capital costs when weighing whether IA/VF farms' produce is considered 'good' food [4,56]. Often, intensive farming is assumed to lead to higher price points because of the capital requirements to build high-tech farms.

A concern noted by Muller [66] and others is that consumers may take some time to adjust to a more affiliative attitude regarding indoor- and other high-tech-grown produce [67]. The emergent literature has identified that consumers may not understand indoor agriculture, but the more they do understand, the more they are supportive of purchasing those products grown in a controlled environment [18]. Most consumers are, in fact, ready to accept produce grown indoors, with a strong preference for this produce if those same consumers are knowledgeable about how the produce was grown. Notably, Seong et al. found that it is increasingly less likely that consumers will reject IA/VF produce if they have more knowledge of how that produce is grown [18].

4.1.5. Scaling

The pursuit of economy-of-scale cost reductions and, therefore, increased profitability continues to plague IA/VFs just as it does other high-tech, capital-intensive industries. While it is perceived as a panacea to many IA/VF challenges by industry players, there is no agreement amongst academics that it is the right time to scale up [43,68].

Profitability goals often hinge on achieving a certain scale of growth regarding the cost of goods sold, and with few mature, successful IA/VF examples, newcomers continue to face the same production challenges [57]. Recent big fund investment in the industry has led start-ups to pursue scale at all costs, which has often caused short-term losses resulting in business failures [58]. If IA/VF operations are scaled more accurately to the needs of immediate markets, they can cater to local needs and see a better contribution margin [14].

Bioeconomic evaluation underlines this strategy by showing that minimizing costs in IA/VF systems is less effective at creating profitability than taking a differentiated product to market and garnering a price premium [14]. CGG produce grown in Australia and exported to Asia is one proven example; creating a stable year-round supply of high-quality, low-contaminant produce that is marketed to a less price-sensitive, affluent consumer base [40] shows how to take a high-quality product to market successfully without achieving scale first.

4.1.6. Overall Assessment of Progress Toward IA/VF Profit Potential

As an overall assessment, the literature has shown that if IA/VF operations are critically managed, from investment flows to controlling the environment and from analytical crop selection to economies of scale, they can be more financially sustainable than many

IA/VF businesses are today. Certain types of IA/VFs can contribute even more to a future food system where demand has shifted to include the specific attributes that vertical farms' controlled environments are perfectly poised to produce. In particular, smaller-scale farms that align capital investment, effective and efficient environmental control, crop selection, high-attribute products, and targeted marketing provide the best current potential for achieving profit.

Table 2 draws upon this discussion to provide a general assessment of IA/VF profitability. Column 1 highlights the most successful IA/VF sector in terms of profitability. Column 2 summarizes the key factors currently impacting progress in profitability. Finally, Column 3 goes beyond the prior analysis to recommend breakthroughs that would have a high probability of moving progress in profitability forward. Each of the three breakthroughs would overcome key concerns with profitability. Forward systems thinking moves the profit analysis to incorporate the economic interactions and tradeoffs across the entire farm operation. Further advances in environmental and cost controls are needed to fully realize the promise of a controlled environment. Artificial intelligence (AI) is needed to achieve control and profitability of the highly complex nature of large-scale farms.

Table 2. Profit sustainability assessment of IA/VF.

3P Components	Current Progress	Breakthroughs Needed
Profit Potential: Smaller-scale farms that align capital investment, effective and efficient environmental control, crop selection, high-attribute products, and targeted marketing provide the best potential for achieving profit.	- Investment opportunities: Tech-focused, enthusiastic investments historically abound, yet without tailored knowledge, this capital can lead to low/negative profits. - High capital needs: High cost of building/maintaining controlled environments creates a financial strain, especially if control systems are resource-optimized. - Scaling difficulties: Pursuit of reduced costs through scale is highly challenging for growing operations; concurrently, large-scale IA/VF farms are vulnerable to managing scale to match market demand and achieve environmental control. - Buyers ready: Consumers are willing to pay for IA/VF produce if the crops selected deliver the attributes demanded. - Market analysis: Matching crop production to market demands quickly is essential in this high-cost structure of agriculture.	- Forward systems thinking: Piecemeal advances in science and practice of IA/VF farms limit ability to make gains in whole system profitability. More models of whole system success, from inputs to marketing, are needed. - Environmental and cost controls: Prioritizing a well-controlled environment, on par with scientifically strong growing standards, might allow any size of IA/VF farm to be profitable. The challenge lies in the lack of knowledge about best practices and the reluctance to share that knowledge across companies. - AI opportunities: Large-scale IA/VFs will require coordination of systems utilizing artificial intelligence with a whole-systems approach matched to large-scale market opportunities.

4.2. Planet: Environmental Impact of IA/VF Agriculture

As with any agricultural business, the impact that IA/VF operations have on their surrounding environments, both upstream and downstream on the value chain, is of utmost importance for ecological sustainability [69]. If we are to integrate more controlled environments into future food systems, it is necessary that IA/VF stakeholders know what comprehensive environmental impact those farms will have, from the building of those

farms to their operations and the lifecycle of the products they sell into the market. The current literature focuses on the carbon emissions of smaller urban farms, not agricultural business entities like most IA/VF operations [75]. The literature also focuses on aspects of localizing supply chains that focus on food miles [76], as well as energy sources and relative energy efficiency of IA/VF farms [59]. More research is needed, specifically the lifecycle analysis of existing IA/VF farms.

To evaluate planet performance, we have approached the analysis of IA/VF's environmental impact from three different angles: the supply chain solutions that IA/VF provides to local markets; IA/VF farms' resource efficiency, especially that of chemicals, water, and land; the well-studied energy impact of IA/VF farms.

4.2.1. Local Supply Chains and the Food Miles Debate

The primary discourse around IA/VF's environmental impact focuses on the subset of farms that are considered urban agriculture: farms growing food in urban areas closer to city-dwelling consumers. When considering the entirety of urban agriculture, a good deal of these farms are rooftop and community-supported gardens, many of which have been found to be carbon-intensive [70], without the technological emphasis of the IA/VF operations discussed in this paper. From this perspective, IA/VF's ability to grow crops that are otherwise very carbon-intensive (e.g., crops like tender herbs that are air freighted from cultivation locations to consumers' geographies) may allow for a segment of IA/VF that can contribute a net reduction in carbon emissions [70].

The evaluation of IA/VF's environmental impact that focuses on a reduction of 'food miles' between the producer and the consumer [77] is one that needs further evaluation. The concept of food miles itself lacks sufficient scientific debate and contains several arguments that could be further clarified. Notwithstanding, some IA/VF operations that have centered their technological focus on reducing the distance between farm and consumer (specifically, in-store IA/VF) have seen positive results in the areas of access, traceability, and reduced carbon emissions of fresh produce, proving that hinging the food miles argument on freshness, quality and food safety (produce performance) is of some value [32]. A broader conversation around supply chain weaknesses and opportunities would begin to unpack the food miles debate, which has been analyzed in the Brazilian produce market [76]. The authors do conclude that, in the Brazilian market, shorter distribution routes (routes with fewer food miles) are often the right choice when evaluated on the criteria of produce performance and demand efficiency (better tailoring supply volumes to consumer preferences), both of which reduce waste and improve logistical efficiency. However, these are not the only factors that determine whether produce supply chains are maximally effective; therefore, there is more to be studied regarding food miles and IA/VF.

4.2.2. Resource Efficiency: Chemicals, Water and Land

Instead, it is likely more useful to evaluate a direct comparison of the environmental impact traditional farming makes versus that of IA/VF. Resource-efficient IA/VF farms reduce the use of resources, including chemicals, by a significant measure compared to similar field-grown crops [44]. Growing indoors also avoids some of the criticisms of traditional agriculture, notably soil degradation, deforestation, and water eutrophication [66]. Definitively, the enclosed nature of IA/VF allows for significant obvious environmental advantages over traditional agriculture when it comes to water efficiency, a critical agricultural resource. IA/VF's systems allow for the recycling of water as opposed to creating runoff and for utilizing fertilizers to almost complete efficiency in these same closed-loop

systems. Moreover, tightly controlled environments in IA/VFs allow for infinitely better food safety due to limited pathogen introduction [4].

Land consideration is an aspect of so many farm decisions, from planting (how many crop cycles could be harvested in a given growing season) to crop selection (what crops the land can sustain in our changing climate) and logistics (how far a grower must drive to deliver produce) to use cases (what the land was used as before, and how that affects agricultural choices). As land that has traditionally been used for agricultural production changes in value, these questions must be considered anew. This includes the trends of moving away from rural land as housing development takes precedence, as well as urban farms reaching the end of an era of inexpensive land in city cores [45]. The research indicates that each urban location's factors will determine the inflection point at which it makes sense to increase the overall quantity of IA/VF farms to answer these growing concerns [28].

The trends of increasing population and urbanization concurrent with unmitigated climate change have resulted in decreasing arable land per person, forcing agriculture to turn away from large rural farms as a single solution to increased demand [40]. In pursuit of necessary yield improvement and a net increase in edible crops, it is imperative that the agricultural industry look straight on at the problem of dwindling arable land. In a future that has ever more demand for fresh food, yet less land for growing that food than ever before, food systems might remain as they are now, with adjustments that allow for more sustainable utilization of soils and surrounding ecosystems. Alternatively, agriculture could take a diversified approach to land use. Utilizing IA/VF's innovative methods would modify food systems in ways that maximize yield per square foot via vertical structures and intense environmental control of all growing inputs and processes [14]. Especially in densely populated urban environments, IA/VFs offer a unique opportunity to augment the supply of fresh food in locales where traditional farming methods simply are not possible, essentially creating arable land in the wake of changing land use.

This choice can be framed as sharing the land in pursuit of more sustainable agricultural production versus sparing the land by diversifying growing methods to include intensive indoor farming [71]. Land sharing implies augmenting the way we have grown food agriculturally in recent history by taking a more sustainable agro-ecological approach. Land-sparing methods use less land to grow crops: farming more intensively, which can be accomplished in myriad ways. To reference land sparing here, we will focus on controlled environment agriculture that allows for growing up instead of out: IA/VF. The acreage utilized in this approach would not have to be traditionally fertile or agronomically appealing [8]. By land-sparing with high-tech farming and intensification, food systems could grow more food with fewer resources in different environments [66]. IA/VF operations have a unique ability to control the factors that contribute to yield, which means they can produce 100 times more land productively than cultivated fields in a comparable region [48]. Both approaches have benefits that could provide for a more sustainable future for modern agriculture, with land-sparing offering advantages such as increased yields and a contracted distance between producer and consumer. Land sharing often fails to incorporate the externalities that traditional farming methods affect in terms of ecological system damages, such as nitrogen and urea runoff, water use, and soil loss [26].

The literature and this analysis lead us to the conclusion that there are myriad resource utilization benefits to augmenting supply chains with IA/VF producers, especially those with dense populations local to the farms themselves. That said, the biggest resource used in IA/VF farms, particularly when considered as farm operational expenses, is that of energy [47].

4.2.3. Energy Use, Sources and Efficiency

Notwithstanding these environmental benefits, there are additional trade-offs between traditional agriculture and IA/VF regarding the value chain environmental impact. Energy remains the biggest cost and one of the biggest environmental impacts due to energy sources' associated carbon footprints. Carbon footprints of vertical farms and greenhouses were reported as high as 5.6–16.7 times and 2.3–3.3 times greater than that of open-field agriculture, respectively [72]. A considerable opportunity to reduce this impact is the installation of sustainable energy-generating equipment (solar, wind, or geothermal) as direct energy sources on IA/VF farmland [73]. Critically, this will involve the participation of local governments in permitting sustainable energy production.

IA/VF's significantly larger energy burden can be grossly attributed to the energy required by grow lights. Paying for each photon of light, which is free as solar energy for plants grown outdoors, adds up both in terms of financial impact as well as environmental impact [73]. Light-driven energy consumption has dropped precipitously in recent years due to light manufacturers' research and development, and there is promise in technological innovation that will continue to draw down energy use needs. While automation, AI (artificial intelligence), and robotics are new applications to IA/VF, they could significantly reduce the labor cost of running farms, from 20% to 30%, while also improving yield, uniformity, and quality of product through tighter controls of operational systems and, therefore, reduced energy use [41,44].

There is more research and discussion to be had about the overall environmental impact of IA/VF operations, both from the built environment and crop production standpoints. A complete lifecycle assessment of existing IA/VF farms would be critical in evaluating the role IA/VF can play in our food system's sustainable response to future demand for fresh produce.

4.2.4. Overall Assessment of IA/VF Progress Toward Planet Potential

The literature has shown that IA/VF operations have significant advantages as well as some considerable challenges when comparing their environmental impact against traditional agriculture. To date, much of the research and discourse around CEA's impact has focused on the concept of food miles, which leaves much to be desired in terms of academically rigorous evaluation. However, those crops that are known to be carbon-intensive and/or are well-positioned to serve the needs of local consumers offer an advantage over the logistics and associated carbon footprint of traditional agriculture. In the same vein, traditional agriculture is often criticized for its environmental impact that centers on inefficient water use, from outsize runoff to incomplete pesticide and fertilizer use; in these areas, IA/VF offers a natural advantage due to its enclosed structures.

Table 3 provides a general assessment of IA/VF planet sustainability impacts. Column 1 highlights the key shortcoming to progress, achieving a promised measurable impact in resource efficiency. Column 2 summarizes the key factors currently impacting progress in planet sustainability. Column 3 recommends the breakthroughs needed to prove IA/VF's net positive planet contributions to future food systems. There is a lack of evaluative research in the areas of lifecycle assessment of IA/VF's built environments, which would complement the body of work on local food supply chains and 'food miles'. Additionally, technological innovation in the area of lighting—a significant energy draw and an area of current focus for many stakeholders in the industry—would continue to aid IA/VF operations in reducing their impact and expenses. Relatedly, investment in renewable energy would augment this energy reality for IA/VF farms.

Table 3. Planet sustainability assessment of IA/VF.

3P Components	Current Progress	Breakthroughs Needed
Planet Potential: Resource efficiency of IA/VF operations is promising, yet the built environment of IA/VF farms has yet to be critically evaluated for its carbon footprint.	- Local wins with reduced food miles: Certain crops with carbon-intensive supply chains have been shown to have a minimized environmental impact when grown close to the end consumer, an opportunity for IA/VF farms' crop selection. - Resource efficiency: IA/VF has been shown to have a much lower impact in the areas of fertilizer and pesticide use, run-off, and water use compared to traditional agriculture. - High energy needs: The resource efficiency of IA/VF farms often pales in comparison to their energy use and their built construction, which are significantly higher than traditional agriculture.	- Lifecycle analysis needed: A lifecycle assessment of IA/VF farms, beyond food miles calculations, would contribute significantly to a critical evaluation of IA/VF's sustainability. - Technological improvements (continued): As lighting is the single biggest financial and environmental impact for these farms, accelerating lighting innovation should continue to edge IA/VF farms into relevance. - Renewables opportunities: Continued investment into renewable energy that feeds IA/VF operations would reduce this environmental impact significantly.

4.3. People: Maximizing Social Benefit

Urban locations for IA/VF farms have been trending as a multi-factorial solution to the trade-offs between high-tech growing and traditional agricultural methods. Many of these potential benefits are social in nature [45]. The existing literature has focused research on the social benefit of IA/VF farms in the following areas: food access in food-insecure areas [63], job creation [40], and resilient supply chains [75,78].

4.3.1. Food Access and Food Insecurity

The social benefit of IA/VF production is often highlighted in locales where food insecurity is of significant concern. Urban farms can provide access to fresh food in food-insecure regions by shortening the value chain between production and consumption. These same farms can be drivers of local employment as well as centers of community, connection, and the distribution of other social 'goods' [79]. They cannot, however, do all those things and remain economically viable in many cases [57]. This brings the policy conversation to the forefront: if IA/VF farms are playing a social benefit role in the communities where they are planted, they should be valued as such by those local governments [75]. Public policy should incorporate a multifunctional benefit analysis for IA/VF operations that exemplify the best of urban agriculture's potential social benefits: urban land re-use, shortened supply chains, the caché of urban farming that lends a high-tech halo to the city permitting such farms; a positive impact on the local community, from agricultural education to job creation [46].

It has been shown that food insecurity exists across rural, peri-urban, and remote communities, in addition to in urban city centers, where food access is a concern alongside that of local food affordability [74]. Rural regions that are typically associated with food production are often faced with a particular type of food insecurity unique to them: food is grown there but inaccessible to those who grow it. In these locations, food access is the primary benefit of IA/VF that the local community would receive.

The challenge for IA/VF to enhance access and reduce food insecurity in urban or rural settings is the current tradeoff between higher-priced, yet higher-quality, IA/VF-grown produce. Tailoring product offerings to price-sensitive consumers puts a strain on the profit

potential of IA/VF operations. Once again, taking a whole systems approach is needed to match potential to reality.

4.3.2. IA/VF Job Creation

Beyond food access, co-locating IA/VF farms near their consumers has the potential for quantifiable social benefits in the form of job creation [45]. CEA jobs span from entry-level agricultural harvesting and packing line jobs to highly technical engineering careers, which appeals to city government decision-makers: a potential boon for start-ups [80]. A recent survey with CEA producers indicated that not only are indoor farms more labor-intensive than greenhouses, but they also employ a significant number of personnel with basic growing skills [83]. In addition, CEA farms can provide intangible community benefits to those stakeholders, such as serving as culturally relevant agricultural education sources [75]. The challenge lies in providing these enriching activities to the local community while also seeking profitability during this tenuous time of industry development.

4.3.3. Supply Chain Resiliency Creates Community Stability

Where the tradeoffs between land use (and cost) abut development concerns, often agricultural regions are driven further and further from population centers, leading to ever-further distances traveled by harvested produce to consumers [45]. This creates a matrix of additional costs, from food waste to fuel surcharges, many of which beg for the contraction of farm-to-consumer logistics and distribution linkages [76]. By building vertical farms in locations where urban infill could be challenging, but population centers are still definitively closer, there is an economic benefit that extends beyond the value those farm products bring to market. This economic benefit, often referred to as the ‘food miles’ solution, has yet to be thoroughly studied for both its planet impact and people impact.

From a supply-chain risk management perspective, resilience involves the system’s capacity to reduce uncertainty in logistics and to prepare for, respond to, and recover from disruptions while sustaining consistent and uninterrupted operations [60,61,81]. In the context of lettuce supply chains in the U.S., its highly perishable nature poses a significant challenge when attempting to accommodate abrupt changes in delivery schedules during major supply chain disruptions. This was observed during the recent COVID-19 pandemic, which exposed vulnerabilities in the supply chain worldwide, contributing to substantial produce wastage [82]. Transportation and distribution, coupled with seasonality in fresh food production, were significant factors disrupting supply chains spanning large geographical distances [60]. IA/VF farms that shorten the distance between producer and consumer in urban environments are a ready solution to the trend of climate change- and pandemic-related supply chain disruptions.

While these theoretical benefits are strong, they are all predicated on having a ready market for the products that IA/VF farms produce [30] and consumers’ readiness to switch from existing farm products to those produced by IA/VF farms nearer to their homes. As discussed in the Profit Section, IA/VF-produced food is still gaining traction in consumers’ minds, leading to a lack of product understanding in some segments [56]. For those consumers who have a positive understanding of IA/VF, the perceived benefits include reduced environmental impacts, decreased food prices, increased job opportunities, a stronger local economy, and the value of food safety as achieved by plant factory growth systems [4]. Consumers can also be somewhat skeptical of IA/VF-grown produce and share the perceived risks of indoor production, including the environmental impact of energy consumption, economic risk due to high startup costs, and NIMBY concerns (conflict with neighbors, smells, and noise). Most importantly, consumers perceived indoor farming products to have a higher price compared to conventional options. This is where the

debate begins regarding IA/VF's potential to contribute to a solution for food insecurity. Accessibility of IA/VF products is a potential concern as well, alongside concerns about equitable distribution of benefits when redeveloping and regentrifying the neighborhoods surrounding these farms [4].

4.3.4. Overall Assessment of IA/VF's Progress Toward People Potential

The potential of IA/VF to contribute positively to the communities in which they are planted is high, though the potential benefits and trade-offs have yet to be thoroughly studied. Notably, the creation of agricultural jobs and knowledge transfer from growers to other local stakeholders is an often unquantified yet significant opportunity. From a logistics perspective, the distance food travels from harvest to plate continues to increase as urban growth drives the cost of land and more agricultural producers, further from the consumers who purchase produce. This trend can further weaken an antiquated supply chain that has been strained to capacity by global pandemics in recent years.

Table 4 provides a general assessment of the impact of IA/VF on people sustainability. Column 1 highlights the fact that the people P lags the research base of the other two P's. Key conceptualizations and measurements are still lacking. Column 2 summarizes key factors currently impacting progress in people sustainability. Column 3 recommends breakthroughs needed to explore, define, and gather empirical data on social impacts. The microeconomic and associated social benefits of IA/VF (which provide education, good jobs, and other associated community benefits) in urban communities need to be thoroughly studied. IA/VF operations are ideally positioned to address areas of food access by shortening supply chains between producers and consumers. This area also needs additional academic research. Supply chain resiliency has been argued as being of utmost importance to local markets, and IA/VF farms can play an active, positive role in shoring up urban food systems. Achieving supply chain benefits needs a more integrated approach to agricultural policy, ideally integrating IA/VF operations to address current and future supply chain challenges. In all three of these areas, fundamental research efforts are needed to achieve breakthroughs.

Table 4. People sustainability assessment of IA/VF.

3P Components	Current Progress	Breakthroughs Needed
People Potential: The people's impact of IA/VF sustainability has probably received less research and practice scrutiny than either profit or planet impacts.	• Social opportunities evident: Social benefits such as job creation, increased food security, and reduced distance between producers and consumers, with myriad benefits, are widely regarded as opportunities for IA/VF farms to capitalize on. • Urban food systems demand growth: Food systems resiliency in an era of major supply chain challenges is augmented by local IA/VF farms located in high-density urban environments. • Unmet policy opportunities: In economically challenged urban locations, there exists a policy opportunity to support IA/VF operations in contributing to the local economy through job creation, education, and space-making; however, this aspect of IA/VF farms has received the least focus in research.	• Microeconomic impact: Additional research on the microeconomic impacts of IA/VF farms on local communities is needed to better incentivize multi-stakeholder collaboration. • Food access: Increased food access due to participation of IA/VF farms needs more research, as merely existing in urban environments does not guarantee better food security; anecdotal evidence shows that communities with IA/VF operations benefit. • Supply chains need policy support: Essential supply chain resiliency can be improved by the presence of IA/VF farms. Policy support of these operations in cities where supply chains are particularly long and/or strained could be beneficial.

5. Discussion

While the IA/VF farm movement is gaining traction, there remain outstanding questions about its impact, as well as the role IA/VF farms, like those mentioned throughout this paper, will play in future food systems. IA/VF's successes and future challenges in the pursuit of profitability are shown in Table 2. The potential microeconomic success, or "P" for profit, of a given IA/VF operation as a business entity is beginning to be understood, but the economic impact of that farm on its immediate community through the creation of jobs, reduction of logistics costs and emissions has yet to be studied.

The environmental impact of IA/VFs (the planet "P"; see Table 3) has been studied in relation to traditional agriculture's known externalities (water use, pesticide/herbicide use, and food safety), yet there are still open questions around the environmental sustainability of IA/VF farms as they are built today. The carbon footprint of IA/VF farms' built environments, analyzed with a lifecycle analysis tool, is one approach that could shed additional light on the debate. While some work has been conducted in this area [62,64], it is ripe for development. The ideal way that IA/VF farms can manage their closed environments, including energy-demanding lights and HVAC, either by tying into the existing grid or through the installation of renewables, still needs further analysis.

Straddling the line between the planet "P" and the people "P" is the concept of food miles, one that is often found in popular media but is not well defined within academia. Specifically, the quantifiable benefit of growing produce closer to the end consumer needs to be critically analyzed. The documented economic benefits (less food waste and product weight loss; less logistics and distribution costs that can be reaped in grower margin or consumer price reductions) and minimized environmental impact (less emissions and fewer externalities) are one part of the story. While some research has begun to uncover the core aspects of food miles [76], further research and analysis to develop a true academic definition of 'food miles' would fortify the policy argument in favor of the establishment of more IA/VF operations. Preliminary research has shown that by intentionally contracting food supply chains, IA/VF can create benefits for various stakeholders along the value chain.

The debate around whether food insecurity is materially improved by the co-location of agricultural producers near urban populations is another unanswered social sustainability question. Urban consumers who have, in recent history, lacked access to fresh food may glean a social benefit from access to CEA farms that extends beyond job creation. The potential drawback of access—whether urbanites who experience food insecurity can purchase these urban-grown farm products—is still debated [4], and this social challenge has yet to be fully understood or addressed by the IA/VF industry.

The social sustainability ('people' P; see Table 4) of IA/VF has some additional unanswered questions. What we do know about the human sustainability of IA/VFs, as well as what we anticipate as necessary advancements, are shown in Table 4. Building an IA/VF farm in any given urban location will, in most cases, represent a net increase in jobs, and those jobs are notably different from traditional agricultural jobs. However, the overall impact of those jobs has yet to be evaluated. Are there technical jobs that will lend IA/VF employees additional skills they can take forward into the job market? Or are IA/VF growers hiring a new version of a field worker who will simply be laboring in a different environment? This question needs to be studied and quantified so city administrators who are considering partnering with IA/VF entities can fully understand the impact of those farms on the microeconomics of the city in question.

6. Conclusions

The analysis and discussion clearly show that CEA, particularly in its IA/VF form, has not fulfilled its sustainability potential for food systems. As to the paper's research question,

the reasons for the lack of fulfillment are complex and many. Within the profit P, only smaller-scale farms have managed to put whole systems together that work in practice. For large-scale ventures, challenges abound: too much capital is needed, there is a limited ability to scale up, only partial ability to fully manage complex internal growing environments, and limited success in developing volume markets for differentiated products. One of the most significant concerns is that profitability analysis is only performed for partial systems and rarely for whole systems. For the planet P, optimizing internal growing environments remains challenging in terms of fully efficient resource use that results in minimal environmental impacts, especially with energy inputs. In addition, distribution and supply chain management need development to minimize the environmental impacts of transportation. The environmental footprint and life cycle analysis are not currently considered, and so-called “food miles” is a weak substitute. The people P is the least well-defined and studied in terms of the IA/VF sustainability impacts. Beyond identifying social impact categories, current literature provides little systematic analysis and conclusions.

Despite the challenges identified, progress has been made on critical components of IA/VF system optimization, particularly in plant science work. More efforts are being made to consider fully robust profitability analyses. Planet and people impacts are increasing part of the analyses being performed in academia and industry. Both public and private partners and partnerships, including those of policymakers, will likely be essential to the full realization of IA/VF’s potential impact on the global food system. The challenges can be met by growing the right crops in the right environment with the most efficient use of resources targeted at consumers at the right place, price, and quality. This combination of factors is always the case in any industry and business, but the risks and rewards are starker for IA/VFs given a growing world population and shrinking resource base for food systems.

7. Future Directions

Future efforts need to focus on solutions to identified challenges. Beyond a summary of current progress, Tables 2–4 also recommend a series of breakthroughs needed to move IA/VF toward fulfilling its sustainability potential. Academia and industry, individually and together, need to create a research and practice agenda to achieve these breakthroughs, including forward systems thinking, more robust environmental and cost controls, harnessing artificial intelligence opportunities, implementing lifecycle analysis, systematic technological improvements, and more comprehensive microeconomic and social impacts on people relying on and working in sustainable food systems. Each breakthrough requires analysis to become more complex and sophisticated than it currently is. More economic modeling of IA/VF systems would support all these breakthroughs. Sharing real performance data between industry and academics would also be critical. Studying the comprehensive scientific and economic causes of specific IA/VF market successes and failures would likewise provide invaluable insights into breakthroughs.

These recommendations for future directions provide a broad starting point. Each needs greater detail as to steps and actions, a valid means to measure progress toward potential, and working partnerships within and between academia, government, and industry.

This paper’s assessment of IA/VF’s progress toward its potential is highly qualitative, given the 3Ps framework for sustainability. The 3P framework allowed for a broad, comprehensive analysis that uncovered causes, roadblocks, and opportunities to IA/VF’s sustainability potential. Other comprehensive frameworks for assessment (e.g., systematic lifecycle analysis) should be used to move forward from this starting point.

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Abbreviations

The following abbreviations are used in this manuscript:

3P	The three P framework: People-Planet-Profit
CEA	Controlled Environment Agriculture
CGG	Clean, Green, and Gourmet
IA	Indoor Agriculture
VF	Vertical Farm

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